

FINANCE CONTROL CENTER

AI-Assisted FP&A & Controlling Portfolio Project

From a spreadsheet-based controlling model to an interactive financial decision-support environment.

Portfolio positioning

A working case study at the intersection of FP&A, Management Control, Business Analysis, data modelling and AI-assisted product development. All financial data shown in the Demo are fully synthetic.

263 Expense transactions	270 Accrual allocations	68/68 Regression tests	0 Data-quality issues
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Reference date: 27 Aug 2026

Synthetic financial data

Live Demo / portfolio URL: [View Live Portfolio](#)

Live Demo: [Launch Live Demo](#)

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Finance Control Center

AI-Assisted FP&A & Controlling Portfolio Project

Project summary

Finance Control Center is an independent portfolio project designed to transform a traditional spreadsheet-based controlling model into a structured, interactive financial management application. The project demonstrates how financial planning and controlling principles can be translated into a usable digital product by combining business requirements, financial logic, data modelling, reporting and AI-assisted development.

The Portfolio Demo uses a fully synthetic dataset and is intended exclusively for demonstration purposes. It does not contain personal financial information or Production data.

The problem

Financial controlling models often start in Excel and gradually evolve into complex workbooks containing transactions, budgets, allocation logic, reconciliations and management reports. While spreadsheets remain highly flexible, increasing complexity can make them harder to maintain, validate and navigate.

The objective of this project was therefore to redesign a multi-layer financial control model into a single interactive environment while preserving the underlying management-control logic.

What the application covers

- Actual vs Budget analysis and variance interpretation.
- Planned commitments and an Operational Outlook based on Actual + explicitly Planned items.
- Cash vs Accrual recognition, including multi-period and fractional cost allocations.
- Recognized Income, Savings Before Investments, Investments and Residual Liquidity.
- Recurring expenses and forward-looking scheduled transactions.
- Business-meal reimbursement tracking and delta analysis.
- Data-quality checks, reconciliation logic and automated regression controls.

August operational outlook

Current-month example - reference date 27 Aug 2026

Actual expenses: €1,316.47 | Planned expenses: €188.49 | Operational Outlook: €1,504.96 | Budget: €1,587.00 |
Variance vs Budget: €82.04 Favorable

This scenario demonstrates how known future commitments can be combined with Actual-to-date performance without presenting the result as a predictive Forecast. Budget remains an independent planning baseline.

Planning and performance logic

Concept	Definition	Management use
Actual	Recorded transactions and recognized financial activity.	Measures realized performance.
Planned	Explicitly scheduled future items with known amounts.	Captures known commitments.
Outlook	Actual + Planned.	Provides an operational view of the expected month/year based only on known items.

Cash vs Accrual

The model separates the original cash payment from the period in which the cost is economically recognized. This allows multi-period expenses to be allocated across months while retaining full traceability to the original transaction.

Example	Cash	Accrual pattern	Control
Professional Online Course	€360	€120 / €120 / €120	Reconciles
Quarterly Sports Membership	€150	€60 / €60 / €30	Reconciles
Annual Insurance	€240	€60 x 4	Reconciles

Data quality and validation

The application includes deterministic controls covering allocation inconsistencies, potential duplicates, orphan allocations, missing categories, invalid dates, Future Actual transactions, recurring-item inconsistencies and accounting-adjustment inconsistencies.

The application also contains automated regression tests covering financial logic, environment isolation, English localization and immutable baseline restoration. The synthetic Demo can therefore be modified during a walkthrough and then reset to the exact original scenario.

Why this matters

The value of the project is not the number of screens or features. It is the ability to translate finance requirements into explicit rules, structure the underlying data, validate the implementation and reconcile the final result.

Role of AI vs my contribution

AI was used as a development accelerator rather than as a substitute for financial judgement. The application does not rely on AI-generated calculations for its core financial logic; the Financial Engine is deterministic.

AI contribution	My contribution
• Accelerated prototyping and implementation. • Generated and revised application code from structured specifications. • Supported iterative debugging and localization.	• Defined the finance requirements and accounting semantics. • Structured the data model and acceptance criteria. • Reviewed AI-generated implementations, identified inconsistencies and drove successive releases. • Validated calculations, reconciliations, data quality and regression behaviour. • Defined project scope, acceptance criteria and final release positioning.

What this project demonstrates

FP&A	Management Control	Business Analysis	Data Modelling
Variance Analysis	Cash vs Accrual	Requirements Engineering	AI-Assisted Development

Disclaimer

All figures, transactions, categories and financial scenarios used in the Portfolio Demo are fully synthetic. The project is intended to demonstrate a working methodology and a practical AI-assisted development workflow, not to provide financial advice or represent a commercial software product.

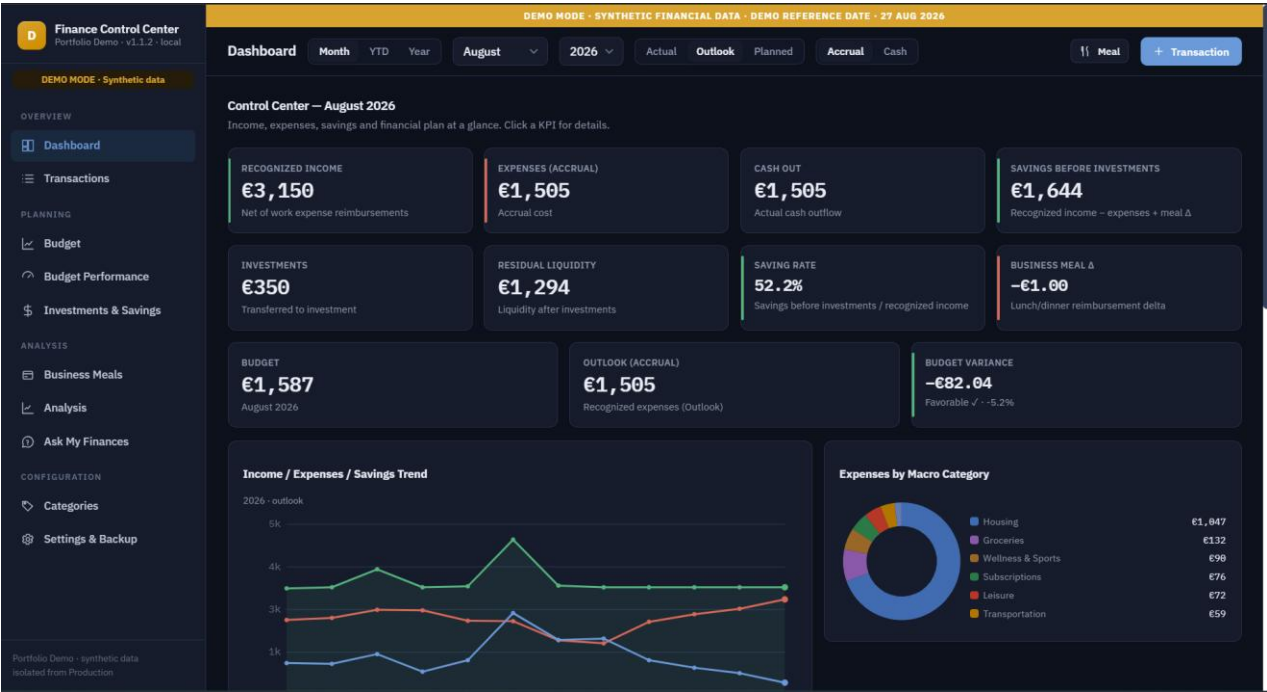
From Spreadsheet Model to Interactive Financial Control Environment

PROBLEM	A multi-sheet Excel controlling model combined transactions, budgets, allocation logic, cash movements and savings metrics. As the model became more layered, it created an opportunity to rethink how the same management-control logic could be structured in an interactive environment.
OBJECTIVE	Build a usable financial-control application that preserves the analytical logic of the original model, improves traceability and distinguishes retrospective Actual performance from known forward commitments without overstating the solution as predictive Forecasting.
APPROACH	Decompose the spreadsheet into business rules and data entities; define canonical models for transactions, allocations, income, Budget, investments, recurring items and reimbursements; separate deterministic Financial Engine logic from presentation; iterate through AI-assisted implementation with explicit acceptance criteria.
SOLUTION	A fully synthetic Portfolio Demo covering Actual vs Budget, Planned commitments, Operational Outlook, Cash vs Accrual, multi-period allocations, recognized income, investments and savings, variance analysis, recurring items, Business Meals, Data Quality and regression controls.
AI VS MY ROLE	AI accelerated prototyping and coding. My role focused on finance semantics, requirements, data-model decisions, acceptance criteria, validation, reconciliation, defect identification and release decisions. The goal was to retain ownership of the business logic while using AI as a technical execution partner.
RESULT	A working case that demonstrates a repeatable method: understand the business logic -> structure the data -> define the rules -> build -> validate -> reconcile -> iterate. The final v1.1.2 EN release is isolated from Production data, resettable to an immutable synthetic baseline and settable to an immutable synthetic baseline.

27 Aug Fixed reference date	€1.50k August Outlook	€82 Favorable vs Budget	68/68 Tests passed
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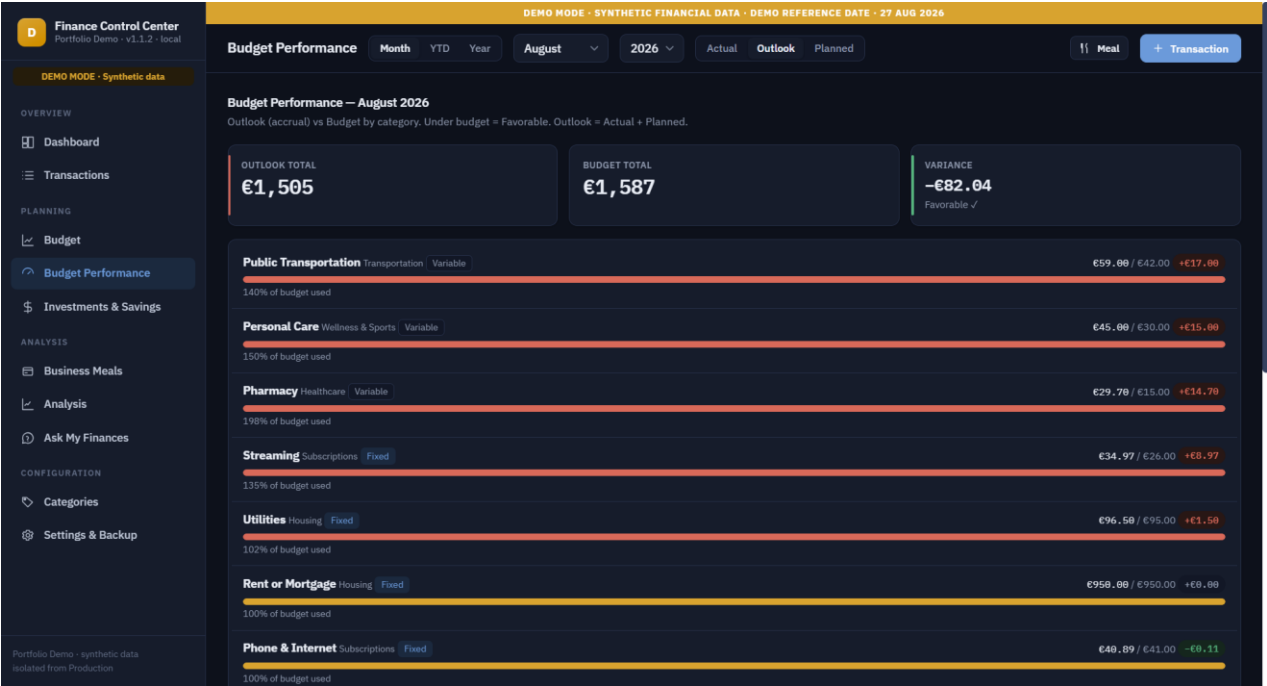
3. SELECTED PRODUCT VIEWS

1. Dashboard - August Operational Outlook



Current-month financial overview combining Actual-to-date and explicitly Planned commitments.

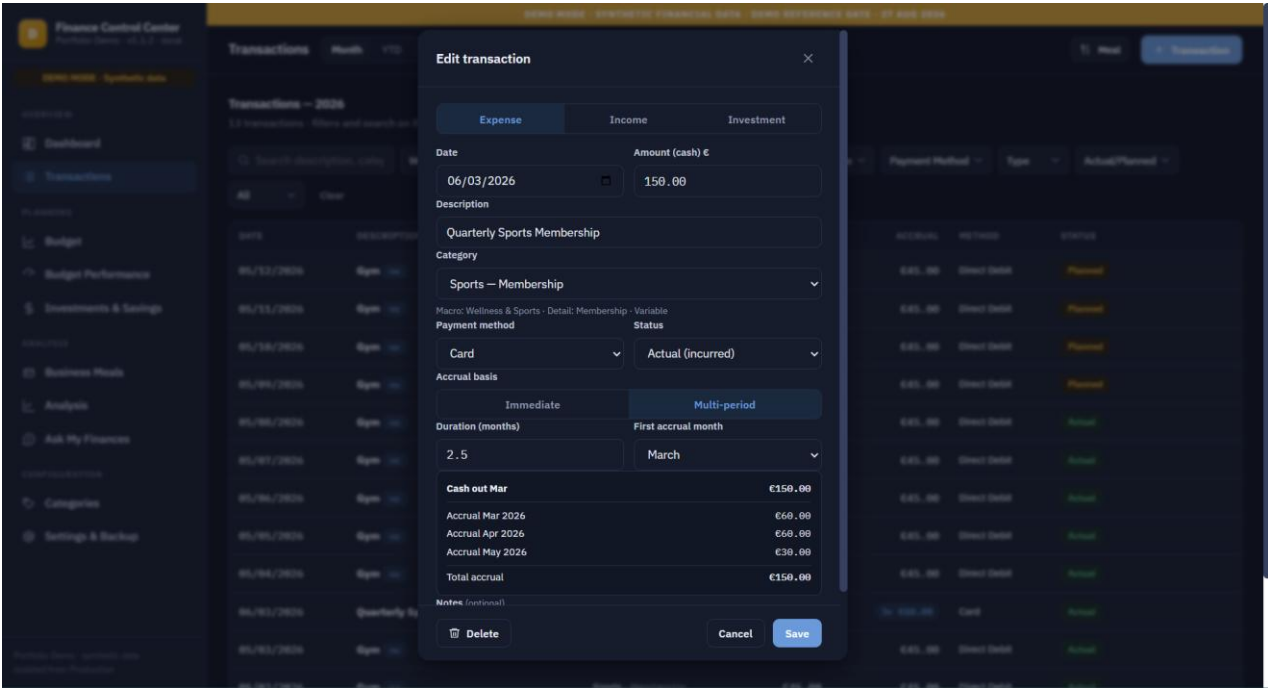
2. Budget Performance - August



Budget variance analysis with drill-down by spending category.

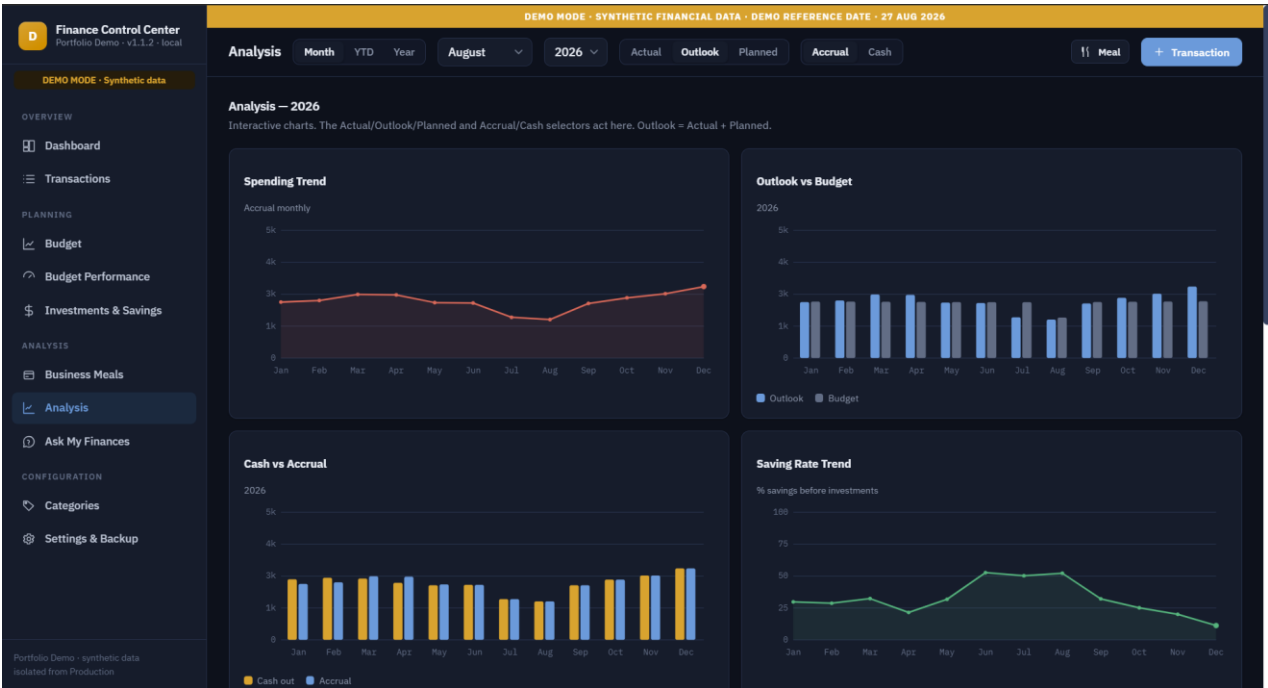
Accounting logic and financial analysis

3. Cash vs Accrual Allocation



Multi-period cost allocation separating cash payment from accrual recognition.

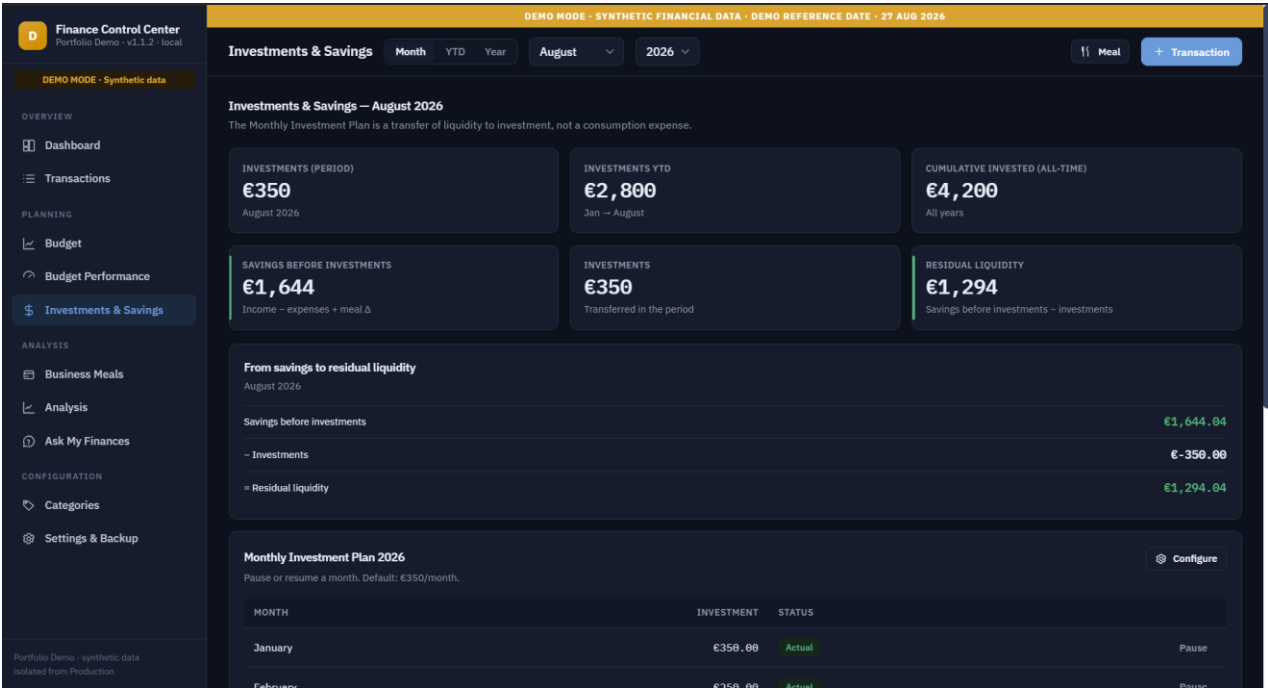
4. Analysis - Monthly Performance



Monthly performance analysis highlighting spending and variance drivers.

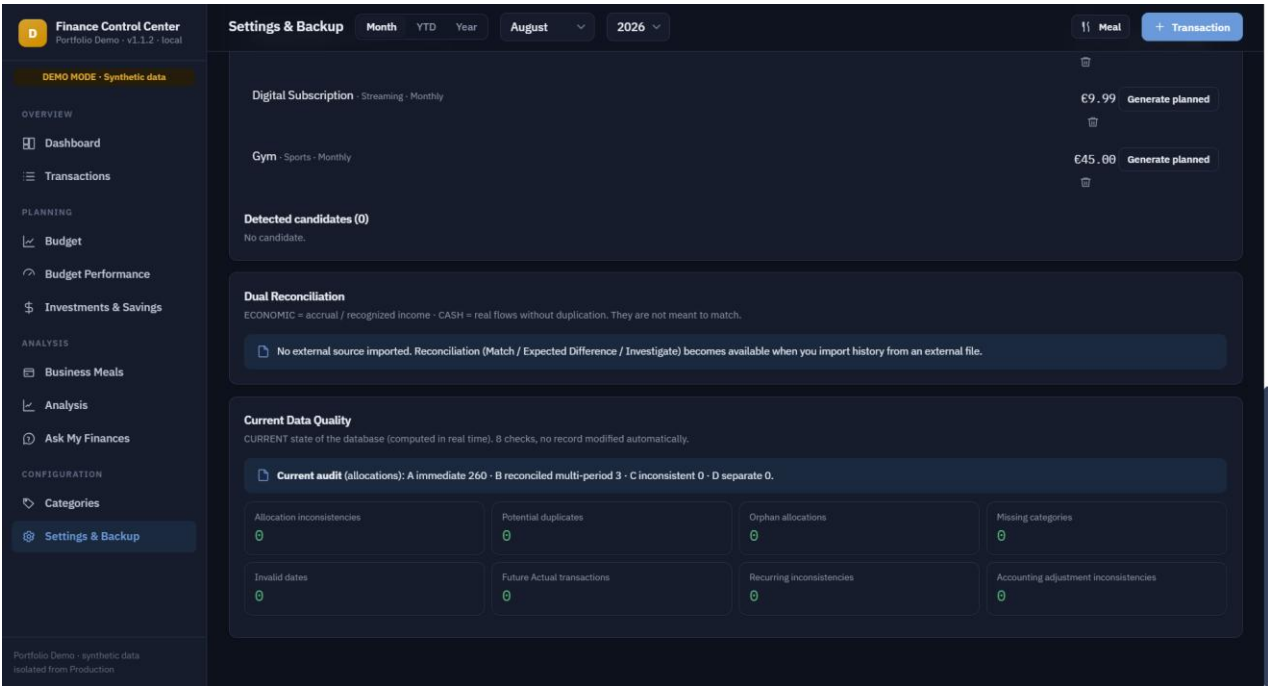
Savings, investments and model reliability

5. Investments & Savings



Integrated savings and investment tracking within the monthly financial model.

6. Data Quality / Automated Tests



Built-in data-quality and regression controls used to validate the financial model.